Track reconstruction: metric reference

In event e, let T_{ei} be the cluster set of eligible MC track i and P_{ej} the cluster set of predicted track j. There are $N_{\mathrm{MC},e}$ eligible tracks. Eligibility retains the existing primary-particle and ownership requirements and at least four owned CVT clusters in the full event. Secondary, decay and shorter tracks are non-target; their clusters remain in the input. Each cluster belongs to at most one truth target and one prediction. Unassigned clusters belong to no P_{ej}.

Pair completeness, purity, IoU and Dice. For a candidate truth–prediction pair,

$$c_{eij} = \frac{|T_{ei} \cap P_{ej}|}{|T_{ei}|}, \qquad p_{eij} = \frac{|T_{ei} \cap P_{ej}|}{|P_{ej}|},$$

$$J_{eij} = \frac{|T_{ei} \cap P_{ej}|}{|T_{ei} \cup P_{ej}|}, \qquad D_{eij} = \frac{2|T_{ei} \cap P_{ej}|}{|T_{ei}| + |P_{ej}|}.$$

Completeness measures the recovered fraction of a truth track; purity measures the fraction of a prediction belonging to that truth track. Prediction size includes all assigned clusters, including other eligible tracks, excluded MC and real-data overlay. Here, a hit denotes a CVT cluster, with unit weight.

Mean track Dice `mc_track_dice_mean`

For each event, Hungarian matching chooses a one-to-one association $\mathcal{M}_e$ maximizing $\sum_{(i,j)\in\mathcal{M}_e} D_{eij}$; zero-overlap pairs are discarded. The dataset score is

$$\overline{D}_{\mathrm{MC}} = \frac{\displaystyle\sum_e \sum_{(i,j)\in\mathcal{M}_e} \frac{2|T_{ei} \cap P_{ej}|}{|T_{ei}| + |P_{ej}|}}{\displaystyle\sum_e N_{\mathrm{MC},e}}.$$

Every eligible truth track has equal weight, with unmatched tracks contributing zero. Only one fragment of a split track can contribute, and a merged prediction can be associated to only one truth track. Independent non-target predictions have no effect on this score.

Global hit completeness and purity `mc_hit_completeness_global / mc_hit_purity_global`

$$C_{\mathrm{MC}} = \frac{\displaystyle\sum_e \sum_{(i,j)\in\mathcal{M}_e} |T_{ei} \cap P_{ej}|}{\displaystyle\sum_e \sum_{i=1}^{N_{\mathrm{MC},e}} |T_{ei}|}, \qquad P_{\mathrm{MC}} = \frac{\displaystyle\sum_e \sum_{(i,j)\in\mathcal{M}_e} |T_{ei} \cap P_{ej}|}{\displaystyle\sum_e \sum_{(i,j)\in\mathcal{M}_e} |P_{ej}|}.$$

These pool cluster counts over the dataset using the Dice association. Completeness includes missed tracks in its denominator; purity includes every cluster in the associated predictions. Thus purity can remain high when many tracks are missed. Total impurity is $1 - P_{\mathrm{MC}}$.

MC hit F1. The harmonic mean of global hit completeness and purity is

$$F_{1,\mathrm{MC}} = \frac{2C_{\mathrm{MC}}P_{\mathrm{MC}}}{C_{\mathrm{MC}} + P_{\mathrm{MC}}} = \frac{2\sum_e \sum_{(i,j)\in\mathcal{M}_e} |T_{ei} \cap P_{ej}|}{\sum_e \sum_{i=1}^{N_{\mathrm{MC},e}} |T_{ei}| + \sum_e \sum_{(i,j)\in\mathcal{M}_e} |P_{ej}|}.$$

For a single pair, $F_{1,eij} = 2p_{eij}c_{eij}/(p_{eij} + c_{eij}) = D_{eij}$: Dice and hit F1 are identical (zero for no overlap). Globally, MC hit F1 pools cluster counts, whereas mean track Dice weights each eligible track equally; they generally differ. This derived count ratio is zero when all eligible tracks are missed, and undefined when its denominator is zero.

Overlay contamination and leakage `mc_bg_contamination_global / bg_leakage_to_mc_global`

Let $R_e = \bigcup_{(i,j)\in\mathcal{M}_e} P_{ej}$ be the associated predicted support, and B_e the set of clusters verified to consist entirely of real-data overlay. Then

$$f_{\text{overlay in MC}} = \frac{\displaystyle\sum_e |R_e \cap B_e|}{\displaystyle\sum_e |R_e|}, \qquad \ell_{\text{overlay}\to\text{MC}} = \frac{\displaystyle\sum_e |R_e \cap B_e|}{\displaystyle\sum_e |B_e|}.$$

The first is the overlay fraction of associated predictions; the second is the fraction of all verified overlay entering those predictions. Excluded MC and wrong-track clusters reduce purity but are not verified overlay. Mixed and MC-present/unresolved provenance are reported separately. Unknown associated provenance makes contamination undefined; unknown provenance anywhere makes leakage undefined. The separately reported contamination lower bound uses only the verified numerator. Any zero-denominator ratio is undefined (JSON null).

Track efficiency and secondary diagnostics

Track efficiency at specified reconstruction quality `track_efficiency_global`

Define the accepted edges by

$$\mathcal{A}_e(p_{\min}, c_{\min}, J_{\min}) = \left\{ (i,j) : \frac{|T_{ei} \cap P_{ej}|}{|P_{ej}|} \geq p_{\min}, \quad \frac{|T_{ei} \cap P_{ej}|}{|T_{ei}|} \geq c_{\min}, \quad \frac{|T_{ei} \cap P_{ej}|}{|T_{ei} \cup P_{ej}|} \geq J_{\min} \right\}.$$

Choose $\mathcal{E}_e \subseteq \mathcal{A}_e$ to maximize the number of one-to-one accepted pairs, then their total IoU. This matching is independent of $\mathcal{M}_e$. The number and fraction of reconstructed eligible MC tracks are

$$N_{\text{reconstructed}} = \sum_e |\mathcal{E}_e|, \qquad \epsilon_{\text{track}}(p_{\min}, c_{\min}; J_{\min}) = \frac{\sum_e |\mathcal{E}_e|}{\sum_e N_{\text{MC},e}}.$$

The standard result uses $p_{\min} = c_{\min} = J_{\min} = 0.5$. For example, $\epsilon_{\text{track}}(0.9, 0.9; 0.5)$ is the fraction of all eligible MC tracks reconstructed with at least 90% completeness and 90% purity. Increasing only $p_{\min}$ measures cleaner reconstruction without requiring more complete recovery. Each set of cuts requires its own accepted matching.

Related quantities. Write $S_e = \bigcup_i T_{ei}$ for eligible support, $U_e = V_e \setminus S_e$ for non-target support, and $A_e = \bigcup_j P_{ej}$ for all assigned clusters; V_e excludes padding. The event-level legacy definitions are

Metric	Equation	Interpretation
Track purity	$\lvert\mathcal{E}_e\rvert / N_{\text{pred},e}$	Accepted fraction of predictions; distinct from cluster purity.
Fake rate / miss rate	$1 - \lvert\mathcal{E}_e\rvert / N_{\text{pred},e}$; $1 - \lvert\mathcal{E}_e\rvert / N_{\text{MC},e}$	Unmatched predictions / unmatched truth tracks. Unmatched non-target predictions may be physical tracks.
Background contamination	$\lvert U_e \cap A_e\rvert / \lvert A_e\rvert$	Non-target fraction of all predictions, including independent non-target tracks.
Background rejection	$\lvert U_e \setminus A_e\rvert / \lvert U_e\rvert$	Non-target clusters left unassigned.
Signal loss to background	$\lvert S_e \setminus A_e\rvert / \lvert S_e\rvert$	Eligible clusters left unassigned.
Split rate / merge rate	$N_{\text{split},e} / N_{\text{MC},e}$; $N_{\text{merged},e} / N_{\text{pred},e}$	Truths overlapping multiple predictions / predictions overlapping multiple truths.

$N_{\text{pred},e}$ counts all nonempty predictions. Splits and merges use any positive overlap. The `matched_*_mean` quantities average pair IoU, purity or completeness over $\mathcal{E}_e$; the field `matched_efficiency_mean` denotes pair completeness.

Adjusted Rand index. For truth–prediction contingency counts n_{ij} on the chosen support, row/column sums $n_{i\cdot}, n_{\cdot j}$ and $N = \sum_{ij} n_{ij}$,

$$\text{ARI} = \frac{\displaystyle\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{n_{i\cdot}}{2} \sum_j \binom{n_{\cdot j}}{2}}{\binom{N}{2}}}{\displaystyle\frac{1}{2}\left[\sum_i \binom{n_{i\cdot}}{2} + \sum_j \binom{n_{\cdot j}}{2}\right] - \frac{\sum_i \binom{n_{i\cdot}}{2} \sum_j \binom{n_{\cdot j}}{2}}{\binom{N}{2}}}.$$

`ari_signal` uses eligible clusters only, excluding contamination outside that support. The legacy single-track convention gives 1 when all its clusters share one prediction label, including unassigned. `ari_with_background` includes non-target truth as one additional class. ARI measures partition agreement.

Aggregation. Historical unsuffixed diagnostics are event means over defined events. Global ratios pool numerators and denominators before division; global matched-quality means pool accepted pairs. Mean track Dice instead weights truth tracks equally. Full definitions, degenerate cases and provenance accounting: `mc_reconstruction_v1.pdf`. All comparisons require the same truth policy, event cohort and prediction settings.